

# AI FOR HEALTH: PREDICTING HEALTH OUTCOMES THROUGH BRFSS AND U.S. HEALTH RANKINGS THROUGH MACHINE LEARNING

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# DATA SOURCES:

| Data source # | Name / short description                              | Source URL                                                                                                                                                                                        | Type:<br>- API<br>- Web page<br>- file | List of fields | Format:<br>- json<br>- xml<br>- csv<br>- sql<br>- other | Have tried to access/collect data with python? yes/no | Estimated data size, number of data points you plan to use |
|---------------|-------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|----------------|---------------------------------------------------------|-------------------------------------------------------|------------------------------------------------------------|
| 1             | BRFSS '23                                             | <a href="https://www.kaggle.com/datasets/isuru-prabath/brfss-2023-csv-dataset?select=BRFSS2023.csv">https://www.kaggle.com/datasets/isuru-prabath/brfss-2023-csv-dataset?select=BRFSS2023.csv</a> | File                                   | Healthcare     | CSV                                                     | Not yet (n)                                           | 300                                                        |
| 2             | Behavioral Risks factors/ CDC Behavioral risk factors | <a href="https://data.cdc.gov/api/v3/views/hn4x-zwk7/query.json">https://data.cdc.gov/api/v3/views/hn4x-zwk7/query.json</a>                                                                       | File                                   | Healthcare     | JSON                                                    | Not yet (n)                                           | 300                                                        |
| 3             | United Healthcare/ America's Health Rankings          | <a href="https://api.americashealthrankings.org/graphql">https://api.americashealthrankings.org/graphql</a>                                                                                       | API                                    | Healthcare     | JSON                                                    | Not yet (n)                                           | 300                                                        |

# INTRODUCTION

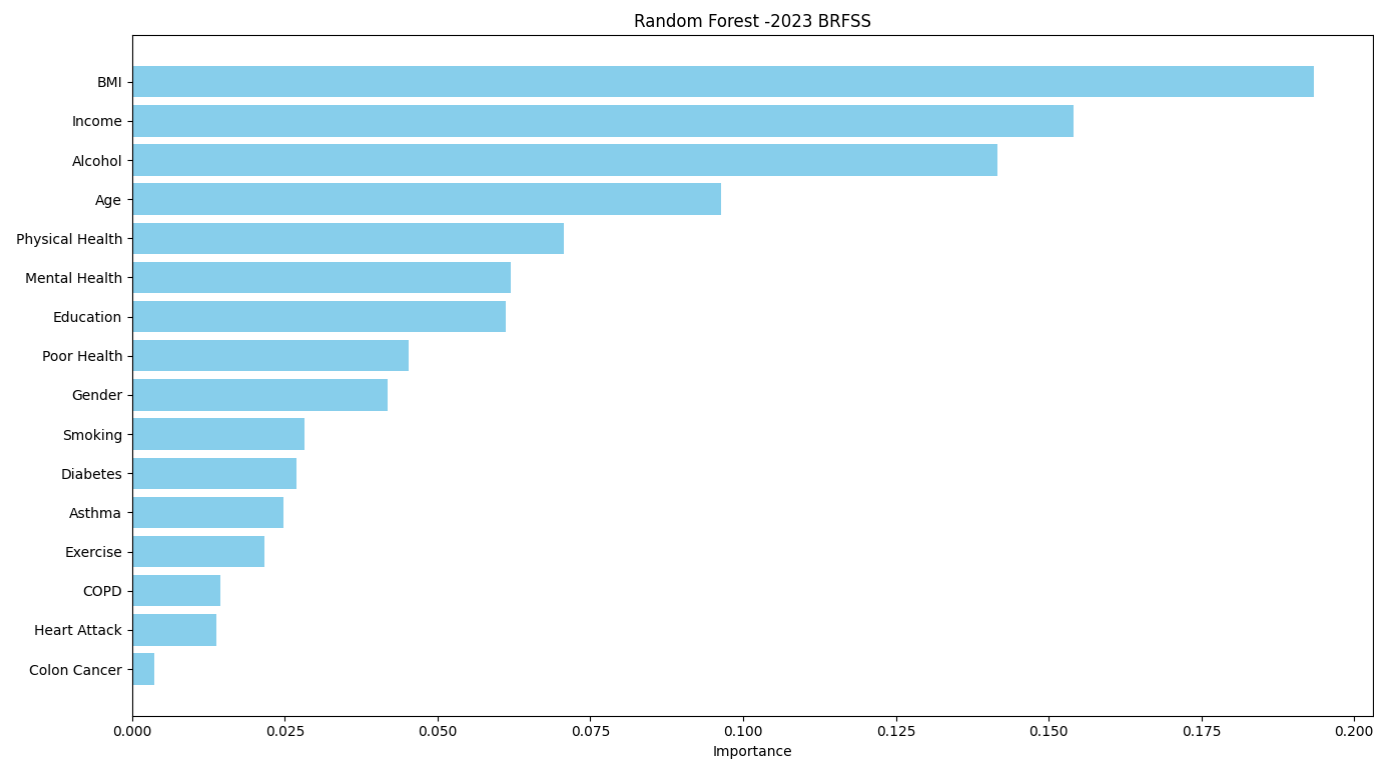
- This project explores how predictive analytics can be used to understand and forecast population health in the United States using data from the 2023 Behavioral Risk Factor Surveillance System (BRFSS). The BRFSS is the nation's largest health-related survey, capturing information on health behaviors, chronic conditions, and access to care. By leveraging both individual-level responses and state-level prevalence data, this project aims to build a model capable of predicting general health outcomes and estimating overall health rankings across all 50 states. The goal is to demonstrate how large-scale public health data can be transformed into meaningful insights that support data driven decision making in the health care industry.



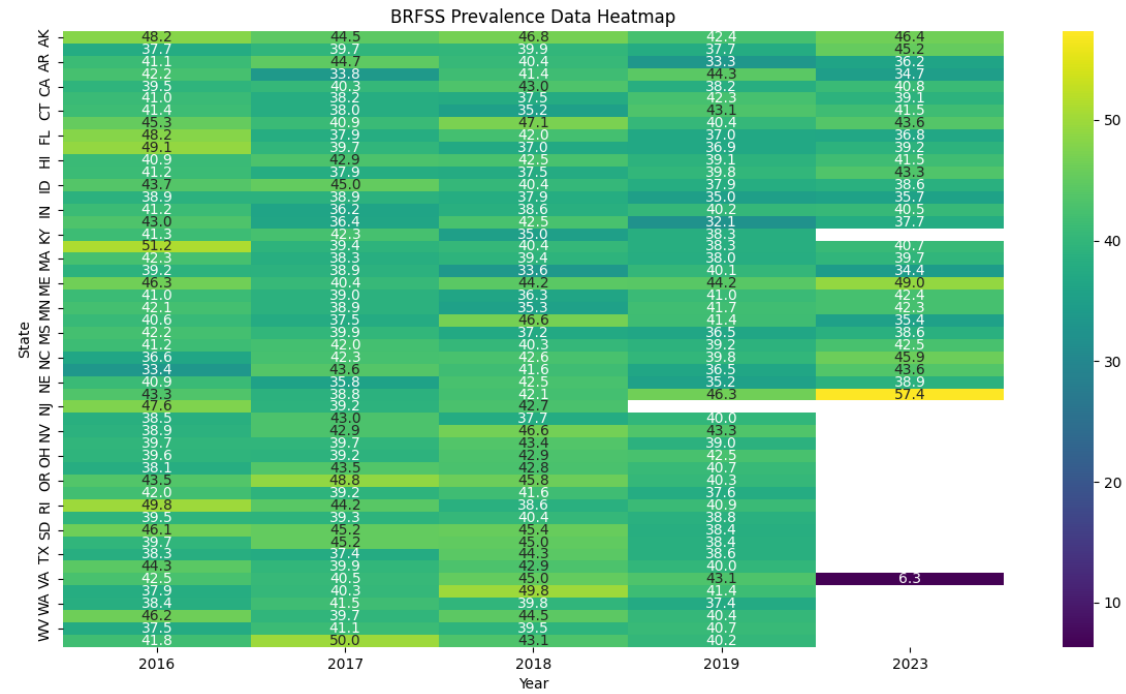
## RESULTS SUMMARY

- Using the 2023 BRFSS data, I built a predictive model to estimate individuals' general health based on key demographic, behavioral, and health-related variables. By incorporating categorical prevalence measures such as obesity rates, smoking status, chronic disease prevalence, and other variables from the BRFSS prevalence dataset, the model captured broader population-level health patterns that influence self-reported health. I then aggregated these predictors to the state level to generate estimates of overall health for all 50 states, producing results that aligned with known U.S. health rankings. Overall, the project demonstrates how BRFSS data can be used to accurately predict both individual general health outcomes and larger state-level health trends.

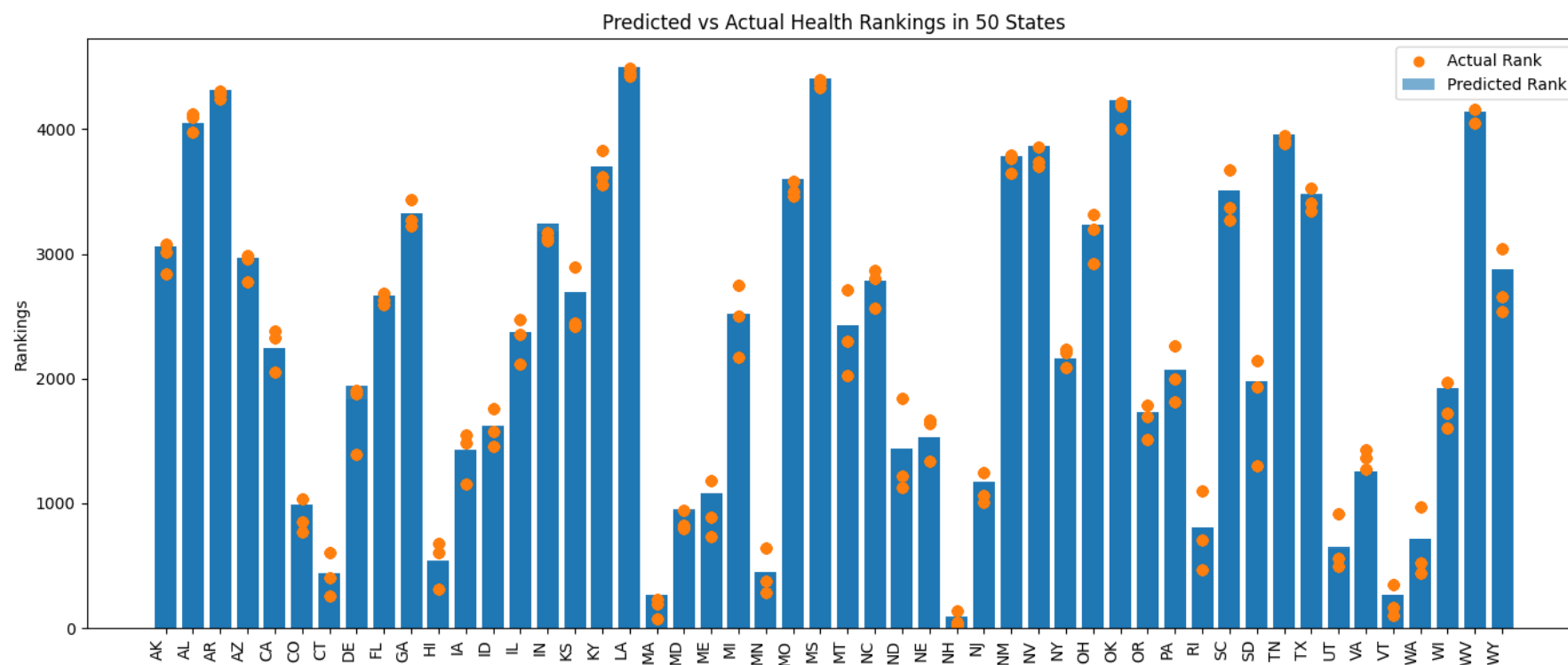
# MACHINE LEARNING MODEL: RANDOM FOREST CLASSIFIER



# BRFSS PREVALENCE DATA HEAT MAP



# ACTUAL VS PREDICTED: HEALTH RANKINGS IN 50 STATES



# CHALLENGES

- A major challenge in this project was working with large datasets in both the 2023 BRFSS dataset and the BRFSS prevalence categorical dataset. Many variables had multiple categories that required careful encoding and cleaning, and a lot of trial and error to obtain the data. Aligning categorical prevalence measures with individual-level data also added complexity. These issues required thoughtful preprocessing and feature selection to keep the model accurate and manageable.







Thank You !